

How Nucleic Acid Synthesis Screening Works—and Why Evasion Is Hard to Stop

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Abstract

Commercial nucleic acid synthesis supports rapid advances in medicine, agriculture, biotechnology, and public health, but it also creates a pathway through which genetic material of concern could be obtained. This report explains how synthesis screening combines automated sequence analysis, customer verification, expert review, regulatory assessment, reporting, and recordkeeping. It examines the technical and institutional factors that make complete prevention of evasion difficult, including biological ambiguity, fragmented procurement, uneven provider participation, limits of identity verification, and advances in artificial intelligence. Drawing on guidance from the International Gene Synthesis Consortium, U.S. government policy, RAND research, SecureBio evaluations, and published benchmarking studies, the report argues for a defense-in-depth model. It recommends common minimum standards, independent performance testing, secure information sharing, supply-chain coverage, safeguards for legitimate scientific access, and adaptive international governance.

Keywords: nucleic acid synthesis; sequence screening; biosecurity; customer verification; defense in depth; artificial intelligence

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1 Introduction

The commercial synthesis of deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) has become an important enabling technology across biomedical research, agriculture, biotechnology, and public health. By allowing researchers to obtain customized genetic material rapidly, synthesis services accelerate legitimate scientific discovery and product development. Nevertheless, the increasing accessibility of these services also creates a dual-use security concern: the same infrastructure could be exploited to obtain genetic material associated with harmful biological agents or functions. Nucleic acid synthesis screening has therefore emerged as a central biosecurity measure intended to identify potentially concerning sequences, customers, and transactions before an order is manufactured or delivered [1, 2].

Screening is not a single technical test but a multistage risk-assessment process. The International Gene Synthesis Consortium (IGSC) recommends both sequence screening and customer screening, supported by record retention, human review, and procedures for resolving or reporting suspicious orders [1]. In practice, automated tools compare requested sequences with databases of regulated pathogens and other sequences of concern, while customer-screening procedures assess identity, institutional affiliation, intended use, and delivery information. These complementary measures reflect a defense-in-depth approach: weaknesses in one layer may be mitigated by additional technical, commercial, and institutional controls. RAND researchers similarly argue that effective governance requires an integrated screening workflow that combines customer verification, order review, information sharing, reporting procedures, and independent conformity assessment [2].

Despite these safeguards, screening remains difficult to implement uniformly. Providers may use different databases, algorithms, thresholds, and escalation criteria, and not all firms participate in voluntary screening regimes. Moreover, fragmented orders, novel or heavily modified sequences, benchtop synthesis platforms, and advances in artificial intelligence (AI) may challenge controls designed around conventional commercial orders. SecureBio’s development of a dedicated DNA-synthesis-screening-evasion evaluation for AI agents illustrates the growing importance of testing screening systems under adversarial conditions [3]. Such evaluations should be used to identify and correct weaknesses, not to disseminate operational methods for circumventing safeguards.

This report explains the principal stages of nucleic acid synthesis screening, examines at a high level why evasion is difficult to prevent completely, and evaluates the limitations of existing safeguards. Drawing on IGSC guidance, SecureBio’s benchmark work, and RAND’s policy analysis, it argues that screening is an essential risk-reduction mechanism but cannot function as an absolute barrier. A resilient governance system must instead combine continuously updated technical screening, robust customer verification, trained human judgment, secure information sharing, conformity assessment, and broad international participation.

2 How Nucleic Acid Synthesis Screening Works

Nucleic acid synthesis screening is best understood as a decision-making workflow rather than a single algorithm. The process begins when a provider receives an order and continues through automated sequence analysis, customer verification, expert review, regulatory assessment, and final disposition. The 2024 U.S. Office of Science and Technology Policy (OSTP) framework organized

this process around six institutional responsibilities: public attestation of adherence, sequence screening, customer screening, reporting of potentially illegitimate orders, retention of screening records, and cybersecurity safeguards for sensitive information [4]. The IGSC protocol translates these broad responsibilities into operating practices for commercial providers [1].

2.1 Order Intake and Initial Data Collection

Effective screening depends on collecting sufficient information before synthesis begins. At a minimum, an order links the requested sequence to a named customer, purchasing organization, shipping destination, and contact information. IGSC members require an institutional name, country, telephone number, email address, and physical shipping address; the protocol also excludes delivery to residential addresses and post-office boxes [1]. These requirements make the transaction more traceable and help the provider determine whether the customer and delivery arrangements are consistent with legitimate scientific or commercial activity.

This stage also establishes the information needed for later review. The provider must be able to associate every ordered sequence with the responsible purchaser and intended recipient, including cases in which an intermediary or distributor is involved. For higher-risk orders, a written description of the intended use may be required. The purpose of collecting this information is not merely administrative: it allows the provider to assess the sequence, customer, institution, destination, and proposed application together rather than treating them as unrelated data points [1, 2].

2.2 Automated Sequence Screening

The submitted nucleic acid sequence is next analyzed against curated data describing regulated pathogens, toxins, and other sequences of concern. Under IGSC Protocol v3.0, providers screen the complete sequence of covered orders against the consortium’s Regulated Pathogen Database and internationally coordinated repositories such as GenBank, EMBL, and DDBJ [1]. Screening both large reference repositories and a curated regulatory dataset serves two different purposes: the former helps identify the biological origin or closest known relatives of a sequence, while the latter connects those results to organisms and genetic elements subject to legal or biosecurity controls.

Screening commonly evaluates both nucleotide and protein-level information. Because different nucleotide strings can encode similar proteins, the IGSC protocol calls for translating a submitted sequence in all six reading frames and comparing the resulting amino-acid sequences with relevant protein records [1]. This approach reduces reliance on exact nucleotide matches and helps identify biologically related material despite ordinary genetic variation or alternative coding choices. Automated analysis nevertheless functions as a filter, not as the final decision-maker. Its purpose is to identify orders that can proceed routinely and those requiring contextual examination by qualified personnel.

The length of sequences included in screening is also significant. IGSC Protocol v3.0 requires screening of complete orders at least 200 base pairs long and calls for members to move toward a 50-base-pair threshold by October 24, 2026, in alignment with the 2024 OSTP framework [1, 4]. Shorter thresholds broaden coverage but also increase computational demands and the number of ambiguous matches requiring interpretation. Consequently, a screening system must balance sensitivity, specificity, processing time, and the burden imposed on expert reviewers.

2.3 Customer and Institutional Verification

Sequence results alone cannot establish whether an order is legitimate. A flagged sequence may be required for vaccine development, diagnostic research, public-health preparedness, or another lawful purpose. Providers therefore conduct customer screening to confirm identity, institutional affiliation, and eligibility to receive the material. The IGSC protocol recommends checking customers against applicable sanctions, denied-party, and export-control lists and performing additional verification when an order involves regulated pathogens, toxins, or other sequences of concern [1].

For an elevated-risk order, the provider evaluates whether the stated purpose is internally coherent and compatible with the customer's organization, expertise, and normal activities. The IGSC emphasizes that this assessment often requires substantial scientific knowledge because the reviewer must understand both the biological significance of the requested material and the plausibility of its proposed use [1]. RAND's analysis similarly treats customer verification as part of a layered system in which technical screening, institutional due diligence, and regulatory compliance inform one another [2]. A sequence match may therefore initiate scrutiny without automatically implying malicious intent.

2.4 Expert Review and Final Disposition

When automated screening identifies a potential sequence of concern, a trained human reviewer examines the match and its context. The reviewer considers the quality and extent of the sequence similarity, the biological function associated with the match, relevant legal controls, the customer's credentials, and the stated purpose of the order. According to the IGSC protocol, the resulting decision may be to accept the order, accept it only after additional customer review, or reject it [1]. This expert stage is essential because reference databases can contain incomplete annotations, closely related harmless organisms, and sequences shared across both regulated and unregulated species.

The decision also incorporates domestic and international legal obligations. Depending on the sequence, customer, and destination, a transaction may implicate select-agent rules, export controls, sanctions, import requirements, or restrictions on particular biological materials. Screening therefore operates partly as a compliance mechanism and partly as a broader biosecurity safeguard. The 2024 OSTP framework sought to make these practices more consistent among providers serving federally funded research, while RAND recommended conformity assessment and shared standards to reduce variation between individual screening programs [4, 2].

2.5 Reporting, Recordkeeping, and Information Security

Orders that remain suspicious after review may be refused and, when appropriate, reported to relevant authorities. IGSC members maintain relationships with law-enforcement and intelligence bodies and reserve the right to share information concerning potentially problematic requests [1]. Reporting decisions require care because customer and sequence data may be commercially sensitive, scientifically valuable, or personally identifiable. Providers must therefore define clear escalation procedures and limit disclosure to circumstances permitted by law and organizational policy.

Recordkeeping makes later auditing and investigation possible. IGSC Protocol v3.0 requires members to retain specified product, delivery, customer, and sequence-screening records for at least eight years [1]. The OSTP framework additionally identifies cybersecurity as a core responsibility because screening systems contain valuable customer identities, proprietary constructs, and information about security controls [4]. Secure storage, access controls, audit trails, and incident-response procedures are thus integral to screening: a poorly protected screening database could itself create privacy, intellectual-property, or biosecurity risks.

Overall, the workflow produces a contextual risk judgment rather than a purely computational answer. Automated tools provide scale and consistency; customer verification supplies institutional context; expert reviewers interpret ambiguous findings; and reporting and recordkeeping support accountability. The effectiveness of the system depends on all of these components functioning together. Screening should consequently be evaluated as a sociotechnical system whose performance is shaped by databases, algorithms, human expertise, organizational incentives, regulatory coverage, and international participation [2, 3].

3 Why Evasion Is Difficult to Prevent Completely

No screening system can guarantee that every concerning order will be identified while every legitimate order proceeds without delay. The central difficulty is that screening must infer biological risk and customer intent from incomplete evidence. A requested sequence may resemble material from a regulated organism yet serve an ordinary research purpose, while a sequence with limited similarity to known references may still have a concerning function. Screening tools must therefore distinguish between biological resemblance, functional risk, regulatory status, and legitimate use—categories that overlap but are not identical [6, 7].

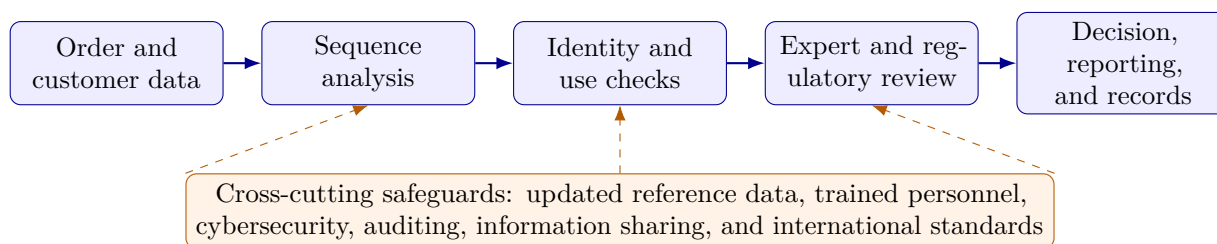


Figure 1: A defense-in-depth screening model. No individual stage is sufficient; reliability emerges from overlapping technical and institutional safeguards.

Figure 1 illustrates why screening is generally treated as a defense-in-depth system. An attempt to avoid one control may still encounter another: sequence analysis may be supplemented by customer verification, expert judgment, regulatory review, or later investigation. Conversely, a weakness shared across several layers can create a systemic gap. The objective is therefore not to construct an infallible filter, but to reduce the probability that a concerning transaction can pass through every layer undetected [2].

3.1 Biological and Computational Ambiguity

The relationship between sequence similarity and biological risk is inherently complex. Closely related organisms may differ significantly in pathogenicity, while many genetic elements are shared by harmful and harmless organisms. Some functions depend on combinations of genes, regulatory context, expression level, host background, or environmental conditions rather than on a single recognizable sequence. As a result, screening based only on whether an order matches a named pathogen can generate both false alarms and missed concerns [7, 6].

Comparative testing confirms that methodological choices matter. A 2025 NIST-led study found that participating screening tools achieved greater than 95% sensitivity and 97% accuracy on a common baseline dataset, but disagreements remained because tools differed in their definitions of a sequence of concern and in their algorithms [5]. These results demonstrate that strong baseline performance is achievable while also showing that no single metric captures every relevant case. Benchmark datasets, shared definitions, and regular external testing are needed to identify inconsistent decisions before they become persistent vulnerabilities.

Reference information also changes over time. New organisms are characterized, gene functions are revised, and regulatory lists are updated at different speeds across jurisdictions. A screening system that is not maintained may gradually lose effectiveness even if its original implementation was technically sound. Providers must consequently update reference databases, software, and review criteria while preserving version histories so that past decisions can be reconstructed during an audit [4, 6].

3.2 Orders Do Not Necessarily Represent Complete Projects

Commercial screening usually evaluates individual transactions, whereas biological projects may involve multiple components, suppliers, institutions, or synthesis platforms. The risk associated with one order can therefore depend on material obtained elsewhere or on information unavailable to the provider. This mismatch between the unit being screened and the larger project limits what any single company can infer from its own records. Earlier analyses identified short products, dispersed procurement, and the growing accessibility of synthesis technologies as reasons that conventional provider-level screening should be reinforced by additional safeguards [8, 2].

The defensive response is not simply to lower a technical threshold. Broader coverage can improve detection, but it may also increase false positives, computational costs, and the volume of cases requiring human review. A sustainable system needs mechanisms for securely linking relevant information across orders while protecting legitimate customers' privacy and intellectual property. Shared alerting procedures, privacy-preserving comparison methods, and clearly defined legal authorities may help providers recognize patterns that would remain invisible within isolated transactions [2].

3.3 Uneven Participation and the Weakest-Link Problem

Screening is most effective when comparable safeguards apply across the market. If responsible providers invest in high-quality review while others apply minimal or no controls, a determined customer may seek the least restrictive route. RAND's assessment emphasized that voluntary

practices vary and that no universally accepted screening standard governed all providers [2]. The problem is international as well as domestic because customers, synthesis firms, distributors, and shipping destinations may be located in different legal jurisdictions.

Uneven adoption also creates an economic problem. Screening requires software, curated data, trained reviewers, secure record systems, and time to resolve uncertain cases. These costs may be proportionally greater for small providers and may place companies that conduct rigorous screening at a competitive disadvantage. Common standards, accessible screening infrastructure, conformity assessment, and procurement rules can reduce this imbalance by making responsible practice a market expectation rather than an optional expense [9, 4].

3.4 Limits of Customer Verification

Customer screening adds information that sequence analysis cannot provide, but identity and intent remain difficult to establish conclusively. Institutional email addresses, professional affiliations, payment records, and shipping information can support a legitimacy assessment without proving the eventual use of the material. Intermediaries and distributors add further distance between a synthesis provider and the final recipient. Reviewers must therefore evaluate whether the available evidence is coherent while avoiding unwarranted assumptions about researchers working in unfamiliar institutions or jurisdictions [1, 2].

This tension creates a persistent trade-off. Verification that is too weak can miss suspicious activity, whereas overly burdensome procedures can delay beneficial research, exclude legitimate customers, and encourage migration to providers outside the screening system. Risk-based review is intended to manage this tension by applying routine processing to low-risk orders and additional scrutiny when sequence, customer, destination, or intended-use indicators justify it [1].

3.5 Artificial Intelligence and Adaptive Pressure

AI adds a new dimension because advanced systems can support biological design, software development, literature analysis, and other tasks that were previously dependent on specialized human expertise. SecureBio’s 2025 work included a dedicated synthesis-screening-evasion evaluation for AI agents, reflecting concern that future models could search for weaknesses more rapidly or systematically than individual users [3]. The significance of such evaluations is defensive: they provide controlled evidence about emerging capabilities and can inform stronger screening algorithms, model safeguards, and monitoring practices.

The challenge is dynamic. Improvements to screening may be followed by new attempts to probe its boundaries, while improvements in biological-design tools may produce sequences that are increasingly difficult to classify using taxonomic similarity alone. Screening programs should therefore incorporate periodic red-team testing, protected evaluation datasets, responsible disclosure, and rapid remediation. Details that would materially enable circumvention should remain access-controlled, while aggregate findings and mitigation lessons should be shared with providers, regulators, and trusted researchers.

3.6 Implications for Defense in Depth

The limits described above do not make screening futile. Rather, they explain why screening should be one component of a broader risk-management architecture. Sequence and customer checks can reduce access through mainstream commercial channels; cybersecurity protects the data and systems supporting those checks; procurement standards reward compliant providers; equipment-level safeguards extend controls to benchtop devices; and public-health surveillance and law enforcement provide additional protection beyond the point of sale [2, 4].

Evasion is difficult to stop completely because the system must operate across changing science, distributed markets, heterogeneous laws, and uncertain human intentions. Its effectiveness should therefore be judged by risk reduction rather than perfection. A mature screening regime will measure performance, investigate failures, revise standards, and coordinate internationally. The strongest strategy is not dependence on a single database or algorithm, but the deliberate construction of overlapping barriers so that failure at one stage does not determine the outcome of the entire system.

4 Strengthening Current Safeguards

The limitations of nucleic acid synthesis screening point toward improvement rather than abandonment. Existing systems already create meaningful friction by requiring sequence analysis, customer verification, and expert review before potentially concerning material is released. The policy challenge is to make these protections more consistent, measurable, and adaptable without imposing disproportionate burdens on legitimate research. A strengthened system should combine common minimum requirements with flexibility for providers to apply additional controls when the characteristics of an order justify closer examination [2, 4].

4.1 Establish a Common and Testable Baseline

The first priority is a clearly defined baseline specifying which providers, products, and transactions must be screened. Common requirements should address sequence and customer screening, reviewer qualifications, escalation procedures, record retention, cybersecurity, and reporting. They should also define the responsibilities of distributors and other intermediaries so that screening obligations cannot be avoided simply by inserting another organization between a customer and a manufacturer. The IGSC protocol and the 2024 OSTP framework provide substantial foundations for such a baseline, but broad effectiveness depends on adoption beyond firms that already participate voluntarily [1, 4].

A standard should be verifiable rather than based solely on self-declaration. Independent conformity assessment could examine whether a provider follows documented procedures, maintains current screening resources, protects sensitive records, and resolves flagged orders appropriately. Certification results need not reveal proprietary algorithms or security-sensitive details; they can instead confirm that a provider meets agreed performance and governance criteria. RAND identifies conformity assessment as an important mechanism for creating confidence across a market in which screening methods and capabilities vary [2].

4.2 Measure Performance Continuously

Screening performance should be evaluated through recurring benchmark exercises rather than a one-time approval. Shared test sets can measure sensitivity, specificity, consistency, and processing burden under controlled conditions. The NIST inter-tool study demonstrates the value of comparing different screening tools against a common dataset: although overall performance was strong, variation in algorithms and definitions produced disagreements that would be difficult to identify from internal testing alone [5]. Regular testing would help providers detect declining performance as databases, biological knowledge, and synthesis products change.

Benchmarks must themselves be governed carefully. Public test cases are useful for transparency and scientific development, while sensitive adversarial cases may require restricted access to prevent evaluation material from becoming a practical guide to circumvention. Results should emphasize aggregate capabilities, recurring failure categories, and corrective actions. SecureBio’s controlled evaluation work offers an example of how emerging AI capabilities can be assessed for defensive purposes while limiting unnecessary disclosure of operational details [3].

4.3 Support Secure Information Sharing

Individual providers possess only partial visibility into the synthesis market. Carefully designed information-sharing mechanisms could help identify repeated suspicious inquiries, coordinated purchasing patterns, or attempts that move between providers after an order is rejected. Such sharing should be limited to information necessary for security purposes and governed by clear rules concerning privacy, commercial confidentiality, data retention, access, and redress. A centralized repository is not the only option; privacy-preserving matching or trusted third-party alerting may provide useful coordination without exposing complete customer records to every participant [2].

Information sharing should also support rapid maintenance of screening resources. Providers need timely notice when regulatory lists, reference annotations, or risk assessments change. Coordinated update mechanisms can reduce the interval during which different firms reach inconsistent decisions because they rely on different versions of the underlying data. Version control and audit records should accompany these updates so that providers can explain which standards and databases informed a particular decision.

4.4 Extend Safeguards Across the Supply Chain

Commercial providers are only one part of the synthesis ecosystem. Distributors, resellers, manufacturers of benchtop synthesis equipment, cloud-based design services, and research institutions may each control information or infrastructure relevant to risk reduction. Responsibilities should therefore be allocated across the supply chain rather than concentrated entirely at the point of commercial manufacture. Equipment-level controls, authenticated access, software updates, and auditable screening functions may be appropriate for benchtop systems, while distributors should preserve the connection between the original customer, requested sequence, and final destination [2, 4].

Procurement policy can reinforce these expectations. Governments, universities, funders, and biotechnology companies can require purchases to be made through providers that follow recog-

nized screening standards. This approach rewards responsible firms and reduces the competitive disadvantage associated with maintaining qualified review staff and secure technical infrastructure. Assistance for smaller providers—including validated tools, training materials, and access to curated reference data—can further reduce the risk that compliance costs push customers toward unscreened markets [9].

4.5 Preserve Legitimate Scientific Access

Security measures should be designed so that they do not unnecessarily obstruct beneficial research. Excessive false positives, opaque decisions, and prolonged reviews can delay diagnostics, vaccine development, and public-health work. Providers should establish clear channels through which customers can supply additional documentation, correct mistaken identity information, and request reconsideration when an order is delayed or refused. Review times and outcomes should be monitored for systematic disparities across institution type, geography, and field of research.

Risk-based review offers a practical balance. Routine orders from verified customers can proceed efficiently, whereas transactions presenting multiple risk indicators can receive more detailed assessment. Decisions should be based on the combined evidence rather than nationality, location, or institutional prestige alone. Transparent high-level criteria, trained reviewers, and documented escalation procedures can improve consistency while preserving the discretion needed to interpret unusual scientific contexts [1].

4.6 Build International and Adaptive Governance

National requirements can improve domestic practice, but synthesis markets and biological research networks operate internationally. Governments, industry groups, standards organizations, and scientific institutions should work toward interoperable minimum expectations for screening, reporting, and data protection. Complete legal uniformity is unlikely; the more realistic goal is mutual confidence that participating jurisdictions apply comparable safeguards and can communicate effectively when a concerning cross-border transaction arises [9, 2].

Governance must also remain adaptive. Standards should include scheduled review periods, mechanisms for urgent revision, and protected channels through which researchers can disclose newly identified weaknesses. Oversight bodies should track changes in synthesis scale, product length, benchtop capabilities, AI tools, and the performance of existing controls. This iterative model treats screening as continuously maintained infrastructure rather than a fixed technical requirement.

Taken together, these measures would create a more resilient screening ecosystem. A common baseline would establish minimum expectations; independent testing would measure whether systems perform as intended; secure information sharing would improve visibility across providers; supply-chain controls would reduce uncovered routes; and international coordination would limit weakest-link effects. None of these measures can eliminate biological risk, but their combined implementation can make failures less frequent, easier to detect, and less consequential.

5 Conclusion

Nucleic acid synthesis screening has become an essential component of modern biosecurity because it places safeguards at a consequential point between digital biological designs and manufactured genetic material. Effective screening does not depend on sequence comparison alone. It combines automated analysis with customer verification, contextual review, regulatory assessment, reporting, recordkeeping, and cybersecurity. The IGSC protocol and the 2024 OSTP framework demonstrate how these elements can be organized into a coherent workflow, while RAND’s analysis shows why they should be treated as mutually reinforcing parts of a defense-in-depth system [1, 4, 2].

The report has also shown why complete prevention of evasion is unrealistic. Biological risk cannot always be inferred directly from taxonomic similarity, and the significance of a sequence may depend on function, combinations with other material, or the context in which it will be used. Individual providers have limited visibility into projects distributed across multiple orders or suppliers. Customer verification can reduce uncertainty but cannot establish intent with certainty, and voluntary adoption creates a weakest-link problem when some firms maintain rigorous programs while others do not. AI and increasingly accessible synthesis technologies add further adaptive pressure to a system that must already respond to changing scientific knowledge [7, 8, 3].

These limitations should not be interpreted as evidence that screening is ineffective. Security controls rarely eliminate risk; their value lies in increasing difficulty, generating warning signals, supporting intervention, and reducing the likelihood that a harmful effort succeeds unnoticed. Benchmark studies indicate that screening tools can achieve strong baseline performance, although continuing differences between tools reinforce the need for shared definitions, independent evaluation, and regular maintenance [5, 6]. Human expertise remains important for interpreting ambiguous findings and distinguishing suspicious transactions from legitimate biomedical and public-health research.

The most credible path forward is therefore a layered and adaptive governance model. Providers should operate under a common, testable baseline; screening tools should be evaluated regularly; relevant information should be shared through secure and legally defined mechanisms; and safeguards should extend to intermediaries and benchtop synthesis systems. Procurement incentives, technical assistance, and international cooperation can broaden participation without placing the full cost of security on individual providers. At the same time, transparent review procedures and appropriate routes for reconsideration are needed to protect equitable access to legitimate scientific services [2, 9].

Ultimately, nucleic acid synthesis screening should be understood as continuously maintained public-interest infrastructure. Its success will depend not on a single perfect algorithm but on the capacity of technical experts, providers, policymakers, funders, and researchers to identify weaknesses and improve the system collectively. As synthesis and biological-design capabilities advance, sustained investment in standards, evaluation, workforce expertise, and international coordination will be necessary to preserve the benefits of synthetic biology while reducing opportunities for misuse.

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